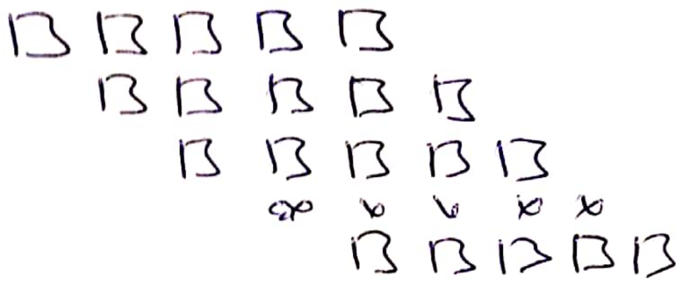
(1.3) Consider the following CPU time diagram

① add S7, S4, S2
② sub S5, S3, S7
③ lw S6, 0(S5)
④ lw S7, 16(S6)



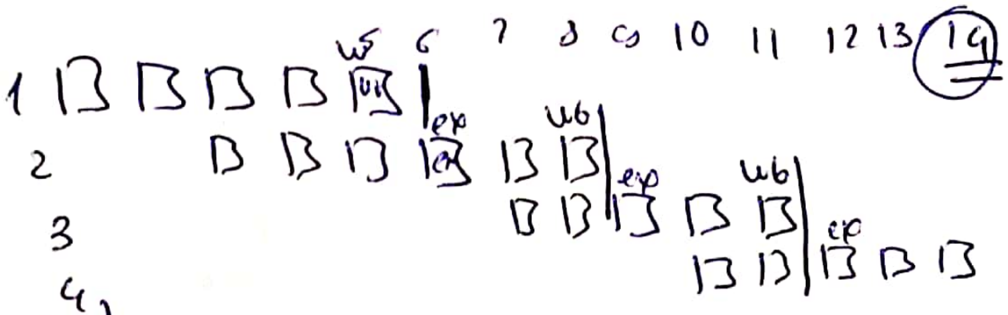
② wich shortcuh one used on each Instruction

①-② alu - alu via S7

①-③ Mem - alu via S5

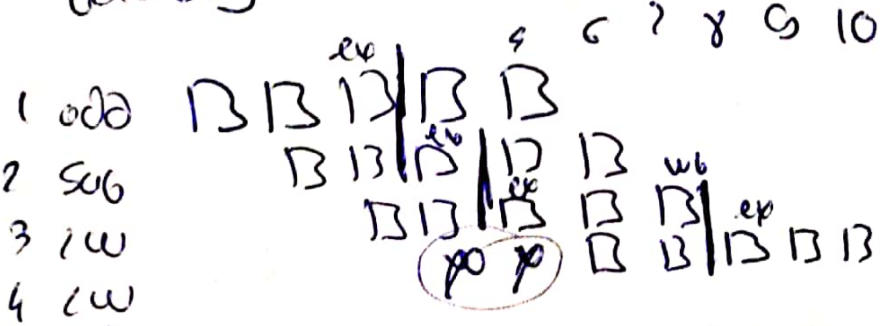
③-④ Mem - alu via S6

c) number of cicles to execute same set of Instrution without forwarding



14 cicles

e) Consider the CPU only Implement alu - alu forwarding



tenho to adde dois nops

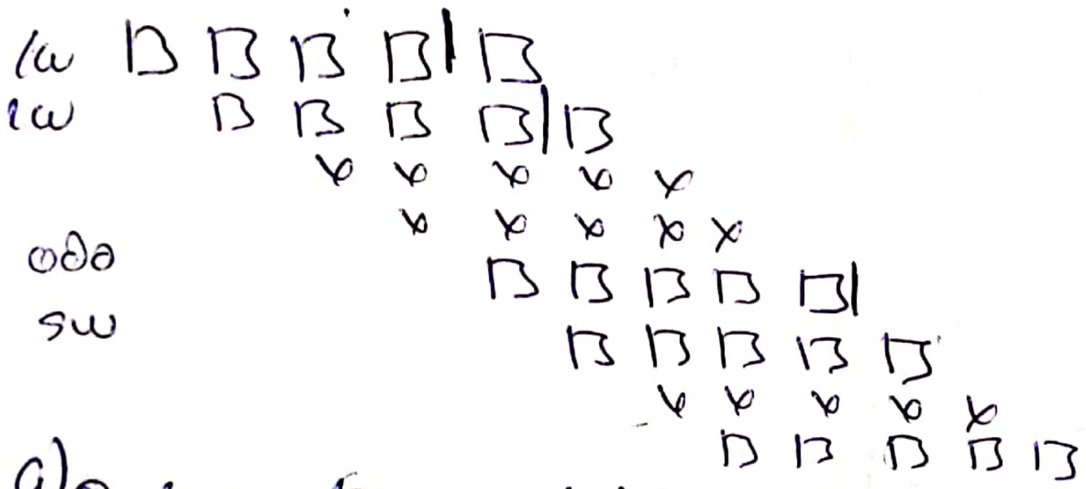
10 cicles alu

14 cicle no forwarding

speedup => $\frac{14 \text{ ciclos}}{10} = 1{,}12$

(5) Sequence of Instruction with CPU without forwarding

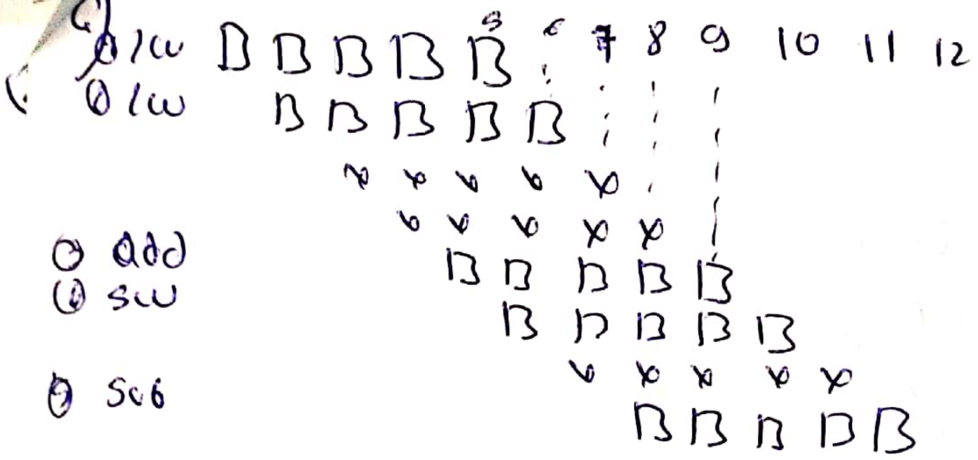
lw r6, 0(r1)
lw —, 4(r1)
add —, —, —
sw —, 4( )
sub —, —, —



a) ① lw r6, 0(r1)
② lw r5, 4(r1) → no dependency
③ add r4, r5, r6 → we need at least 2 nop for 2 instruction to get r5 register
④ sw r5, 4(r5)
⑤ sub r0, r4, r5

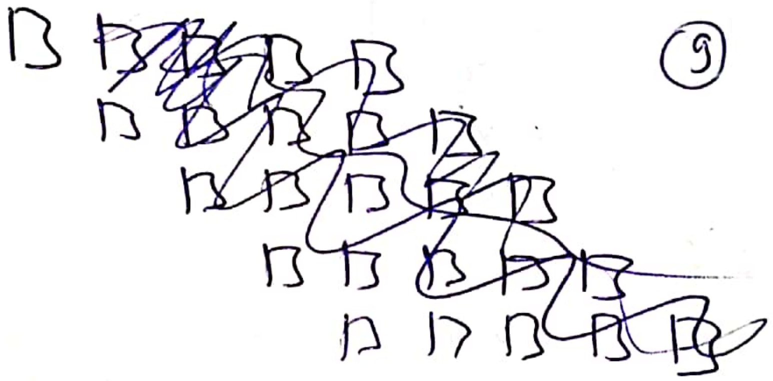
↳ all ok

↳ have a dependency from r4 to sub where we require 2 nop



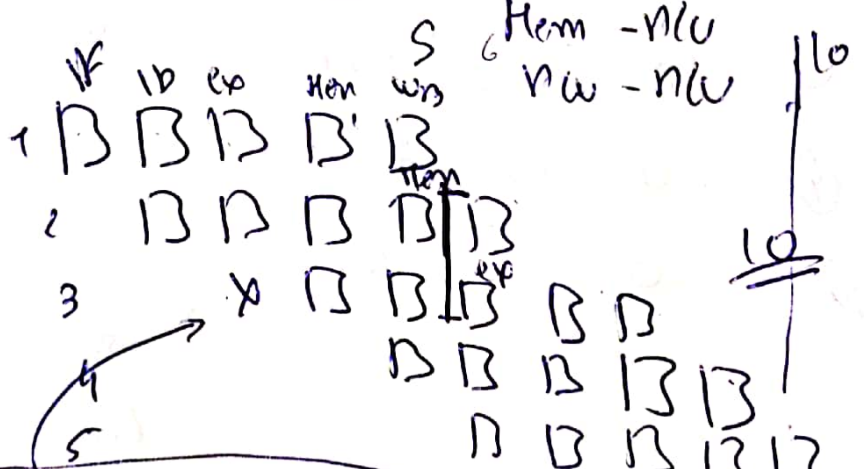
① ~~add~~ → Instruction 3 (add) is in Write Back
→ Instruction 4 (sw) is in Mem
→ Instruction 5 (sub) is in ID

c)



⑨

d) $\frac{12}{9} = \frac{4}{3}$



① lw t6, 0(t1)
② lw t5, 4(t1)
③ add t4, t5, t6
④ sw t5, 0(t5)
⑤ sub t0, t4, t5

Mem-ALU + reordo

②

```
        sub   t0, t0, t0                t1 = 500
        addi  t3, zero, 50
loop:
        lw    t2, 0(a0)      1
                                 +2
        add   t0, t2, t2     4
        addi  a0, a0, 4      5
                                 +2
        addi  t3, t3, -1     8
                                 +2
        bne   t3, zero, loop 11
        add   a0, zero, t0   13
```

12 x 50 +3

607

b)

```
        sub   t0, t0, t0
        addi  t3, zero, 50
loop:
        lw    t2, 0(a0)   1
        ~~addi~~ xxxxx
        add   t0, t2, t2  3
        add   a0, a0, 4   4
        addi  t3, t1, -1  5
        bne   t3, zero, loop 6
        add   a0, zero, t0
```

6 x 50 + 3 = ____

[illegible] mod
calculate

CS CamScanner